

BradleyGastaldiTerilla2023: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
BradleyGastaldiTerilla2023_F00001	2	1.000	X	X	When considering a mathematical object X that has little
BradleyGastaldiTerilla2023_F00002	2	1.000	$X \rightarrow \operatorname{Fun}(X)$	$X \rightarrow \operatorname{Fun}(X)$	$X \rightarrow \operatorname{Fun}(X)$ so that when working with $\operatorname{Fun}(X)$, one is
BradleyGastaldiTerilla2023_F00003	2	1.000	$\operatorname{Fun}(X)$	$\operatorname{Fun}(X)$	$X \rightarrow \operatorname{Fun}(X)$ so that when working with $\operatorname{Fun}(X)$, one is
BradleyGastaldiTerilla2023_F00004	2	1.000	k^X	k^X	The first example that comes to mind is k^X , the set of
BradleyGastaldiTerilla2023_F00005	2	1.000	k	k	functions on a set X valued in a field k , which forms a vec-
BradleyGastaldiTerilla2023_F00006	2	1.000	$X \rightarrow k^X$	$X \rightarrow k^X$	tor space. The embedding $X \rightarrow k^X$ is defined by sending
BradleyGastaldiTerilla2023_F00007	2	1.000	$x \in X$	$x \in X$	$x \in X$ to the indicator function of x defined by $y \mapsto \delta_{xy}$.
BradleyGastaldiTerilla2023_F00008	2	1.000	x	x	$x \in X$ to the indicator function of x defined by $y \mapsto \delta_{xy}$.
BradleyGastaldiTerilla2023_F00009	2	1.000	$y \mapsto \delta_{xy}$	$y \mapsto \delta_{xy}$	$x \in X$ to the indicator function of x defined by $y \mapsto \delta_{xy}$.
BradleyGastaldiTerilla2023_F00010	2	1.000	$\langle f \mid g \rangle = \sum_{x \in X} f(x) g(x)$	$\langle f \mid g \rangle = \sum_{x \in X} f(x) g(x)$	ned by $\langle f \mid g \rangle = \sum_{x \in X} f(x) g(x)$ making k^X into a Hilbert
BradleyGastaldiTerilla2023_F00011	2	1.000	$ x\rangle$	$ x\rangle$	space. The physics “ket” notation $ x\rangle$ for the indicator func-
BradleyGastaldiTerilla2023_F00012	2	1.000	$ x\rangle \in k^X$	$ x\rangle \in k^X$	the vector $ x\rangle \in k^X$ and reminds us that there is an inner
BradleyGastaldiTerilla2023_F00013	2	1.000	$\{ x\rangle : x \in X\}$	$\{ x\rangle : x \in X\}$	vectors $\{ x\rangle : x \in X\}$ become an ordered basis for k^X and
BradleyGastaldiTerilla2023_F00014	2	0.993	$\{0, 1\}$	$\{0, 1\}$	tum information theory. Classical bits $\{0, 1\}$ are replaced
BradleyGastaldiTerilla2023_F00015	2	1.000	$\{ 0\rangle, 1\rangle\}$	$\{ 0\rangle, 1\rangle\}$	by quantum bits $\{ 0\rangle, 1\rangle\}$ which comprise an orthonormal
BradleyGastaldiTerilla2023_F00016	2	1.000	$\mathbb{C}^{\{0, 1\}}$	$\mathbb{C}^{\{0, 1\}}$	basis of a two-dimensional complex Hilbert space $\mathbb{C}^{\{0, 1\}}$.
BradleyGastaldiTerilla2023_F00017	2	1.000	$x \neq y$	$x \neq y$	if $x \neq y$, then the sum $ x\rangle + y\rangle$ will have all zeroes with 1s
BradleyGastaldiTerilla2023_F00018	2	1.000	$ x\rangle + y\rangle$	$ x\rangle + y\rangle$	if $x \neq y$, then the sum $ x\rangle + y\rangle$ will have all zeroes with 1s
BradleyGastaldiTerilla2023_F00019	2	1.000	y	y	in both the x - and y - entries and the sum $ x\rangle + x\rangle$ has all
BradleyGastaldiTerilla2023_F00020	2	1.000	$ x\rangle + x\rangle$	$ x\rangle + x\rangle$	in both the x - and y - entries and the sum $ x\rangle + x\rangle$ has all
BradleyGastaldiTerilla2023_F00021	2	1.000	$k = \{0, 1\}$	$k = \{0, 1\}$	$k = \{0, 1\}$, the vector space structure seems a little weak.
BradleyGastaldiTerilla2023_F00022	2	1.000	$\{0, 1\}^X$	$\{0, 1\}^X$	structures on $\{0, 1\}^X$. Elements of $\{0, 1\}^X$ can be thought of

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BradleyGastaldiTerilla2022 F00023	2	■ 1.000	$v \vee w$	$v \vee w$	join $v \vee w$ and the meet $v \wedge w$ of two vectors correspond to
BradleyGastaldiTerilla2022 F00024	2	■ 1.000	$v \wedge w$	$v \wedge w$	join $v \vee w$ and the meet $v \wedge w$ of two vectors correspond to
BradleyGastaldiTerilla2022 F00025	2	■ 1.000	v	v	v and w , and neither the meet nor the join coincide with
BradleyGastaldiTerilla2022 F00026	2	■ 1.000	w	w	v and w , and neither the meet nor the join coincide with
BradleyGastaldiTerilla2022 F00027	2	■ 0.998	$0 \leftrightsquigarrow 1$	$0 \leftrightarrow 1$	by interchanging $0 \leftrightarrow 1$, the vectors in $\{0, 1\}^X$ are partially
BradleyGastaldiTerilla2022 F00028	2	■ 0.244	$\{ \}^{\text{CP}}$	CP	theory. By replacing a category $\mathcal{C}$ by Set^{cop} , the set-valued
BradleyGastaldiTerilla2022 F00029	2	■ 0.908	F	F	for accuracy. It's common to call a functor F in Set^{cop} a
BradleyGastaldiTerilla2022 F00030	2	■ 0.908	$\{ \}^{\text{cop}}$	cop	for accuracy. It's common to call a functor F in Set^{cop} a
BradleyGastaldiTerilla2022 F00031	2	■ 0.748	$\{ \}^{\text{COP}}$	COP	$\mathcal{C} \rightarrow \text{Set}^{\text{cop}}$ of the original category as a full subcategory
BradleyGastaldiTerilla2022 F00032	2	■ 0.290	$\{ \}^{\text{c op}}$	c op	of Set^{cop} . Given an object x in $\mathcal{C}$, Grothendieck used h^x
BradleyGastaldiTerilla2022 F00033	2	■ 0.290	$h^{\{x\}}$	h^x	of Set^{cop} . Given an object x in $\mathcal{C}$, Grothendieck used h^x
BradleyGastaldiTerilla2022 F00034	2	■ 1.000	$h^{\{x\}} := \text{C}(-, x)$	$h^x := \text{C}(-, x)$	to denote a <i>representable presheaf</i> $h^x := \text{C}(-, x)$ which is
BradleyGastaldiTerilla2022 F00035	2	■ 1.000	$\text{C}(y, x)$	$\text{C}(y, x)$	defined by mapping an object y to the set $\text{C}(y, x)$ of mor-
BradleyGastaldiTerilla2022 F00036	2	■ 0.883	$x \mapsto h^{\{x\}}$	$x \mapsto h^x$	by $x \mapsto h^x$. And just as the vector space k^X has more struc-
BradleyGastaldiTerilla2022 F00037	2	■ 0.388	$\{ \}^{\text{Cop}}$	Cop	ture than X , the category Set^{cop} has more structure than $\mathcal{C}$.
BradleyGastaldiTerilla2022 F00038	2	■ 0.930	$\text{X} = \text{X}^{\text{op}}$	$\text{X} = \text{X}^{\text{op}}$	$\text{X} = \text{X}^{\text{op}}$, and a presheaf F on X assigns a set to every ob-
BradleyGastaldiTerilla2022 F00039	2	■ 0.988	$F(x)$	$F(x)$	with the set $F(x)$ in the x -entry. The representable func-
BradleyGastaldiTerilla2022 F00040	2	■ 0.996	$\emptyset$	$\emptyset$	x -entry. Using 0 for $\emptyset$ and 1 for $*$ produces the same ar-
BradleyGastaldiTerilla2022 F00041	2	■ 0.996	$x \amalg y$	$x \amalg y$	k^X . Notably, the categorical coproduct $x \amalg y$ does not ex-
BradleyGastaldiTerilla2022 F00042	2	■ 0.786	$h^{\{x\}} \coprod h^{\{y\}}$	$h^x \amalg h^y$	ist in the category X , but the coproduct $h^x \amalg h^y$ of the rep-
BradleyGastaldiTerilla2022 F00043	2	■ 1.000	$h^{\{y\}}$	h^y	resentable functors h^x and h^y <i>does</i> exist in the category of
BradleyGastaldiTerilla2022 F00044	2	■ 0.908	$*$	$*$	ing of empty sets except for a one-point set $*$ in the x - and
BradleyGastaldiTerilla2022 F00045	2	■ 0.551	$h^{\{x\}} \coprod h^{\{x\}}$	$h^x \amalg h^x$	y -entries; the coproduct $h^x \amalg h^x$ consists of all empty sets
BradleyGastaldiTerilla2022 F00046	2	■ 0.993	$* \sqcup *$	$* \sqcup *$	except for a two point set $* \sqcup *$ in the x -entry. And just as
BradleyGastaldiTerilla2022 F00047	2	■ 0.998	$\text{X}^{\text{op}} \rightarrow$	$\text{X}^{\text{op}} \rightarrow$	every functor $\text{X}^{\text{op}} \rightarrow \text{Set}$ is constructed from representable
BradleyGastaldiTerilla2022 F00048	2	■ 0.731	$\{ \}^{\text{Xop}}$	Xop	presheaf in Set^{Xop} is a colimit of representables.

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BradleyGastaldiTerilla2022 F00049	<u>2</u>	■ 0.998	$0 \rightarrow 1$	$0 \rightarrow 1$	and one nonidentity morphism $0 \rightarrow 1$, results in the cate-
BradleyGastaldiTerilla2022 F00050	<u>2</u>	■ 0.701	$2^{\{\text{CP}\}}$	2^{CP}	gory $2^{\text{C}^{\text{op}}}$ of 2-valued presheaves on $\mathbf{C}$. In the case when $\mathbf{C}$ is
BradleyGastaldiTerilla2023 F00051	<u>3</u>	■ 0.998	$2^{\{\mathrm{X}\}}$	$2^{\mathbf{X}}$	in $2^{\mathbf{X}}$ are exactly the same as $\{0, 1\}$ -valued functions on X ,
BradleyGastaldiTerilla2023 F00052	<u>3</u>	■ 1.000	f	f	presheaves f and g is the join (union) $f \vee g$, and the cate-
BradleyGastaldiTerilla2023 F00053	<u>3</u>	■ 1.000	g	g	presheaves f and g is the join (union) $f \vee g$, and the cate-
BradleyGastaldiTerilla2023 F00054	<u>3</u>	■ 1.000	$f \vee g$	$f \vee g$	presheaves f and g is the join (union) $f \vee g$, and the cate-
BradleyGastaldiTerilla2023 F00055	<u>3</u>	■ 1.000	$f \wedge g$	$f \wedge g$	gorical product is the meet (intersection) $f \wedge g$. So for any
BradleyGastaldiTerilla2023 F00056	<u>3</u>	■ 1.000	$F_2 = \{0, 1\}$	$F_2 = \{0, 1\}$	vector space over $F_2 = \{0, 1\}$, the field with two elements,
BradleyGastaldiTerilla2023 F00057	<u>3</u>	■ 0.952	$2 = \{0, 1\}$	$2 = \{0, 1\}$	or as enriched presheaves on $\mathbf{X}$ valued in $2 = \{0, 1\}$, depend-
BradleyGastaldiTerilla2023 F00058	<u>3</u>	■ 0.512	i n	in	ture of what the functions are valued <i>in</i> . For example, the
BradleyGastaldiTerilla2023 F00059	<u>3</u>	■ 1.000	G	G	other example, passing from a finite group G to the free vec-
BradleyGastaldiTerilla2023 F00060	<u>3</u>	■ 1.000	$\mathbb{C}^G$	$\mathbb{C}^G$	tor space $\mathbb{C}^G$ doesn’t tell you much about the group, until
BradleyGastaldiTerilla2023 F00061	<u>3</u>	■ 1.000	$\mathbb{C}^G \rightarrow \mathbb{C}^G$	$\mathbb{C}^G \rightarrow \mathbb{C}^G$	$\mathbb{C}^G \rightarrow \mathbb{C}^G$. The result is the regular representation for G ,
BradleyGastaldiTerilla2023 F00062	<u>3</u>	■ 0.999	$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$	form, a DNN can be described as a function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$
BradleyGastaldiTerilla2023 F00063	<u>3</u>	■ 1.000	$\mathbf{M}_i$	$\mathbf{M}_i$	where the $\mathbf{M}_i$ are $n_i \times n_{i-1}$ matrices, the $b_i \in \mathbb{R}^{n_i}$ are “bi-
BradleyGastaldiTerilla2023 F00064	<u>3</u>	■ 1.000	$n_i \times n_{i-1}$	$n_i \times n_{i-1}$	where the $\mathbf{M}_i$ are $n_i \times n_{i-1}$ matrices, the $b_i \in \mathbb{R}^{n_i}$ are “bi-
BradleyGastaldiTerilla2023 F00065	<u>3</u>	■ 1.000	$b_i \in \mathbb{R}^{n_i}$	$b_i \in \mathbb{R}^{n_i}$	where the $\mathbf{M}_i$ are $n_i \times n_{i-1}$ matrices, the $b_i \in \mathbb{R}^{n_i}$ are “bi-
BradleyGastaldiTerilla2023 F00066	<u>3</u>	■ 1.000	a	a	ases”, the a is a (nonlinear) “activation” function, and g
BradleyGastaldiTerilla2023 F00067	<u>3</u>	■ 1.000	M_i	M_i	parametrized by the entries of the matrices M_i and the bias
BradleyGastaldiTerilla2023 F00068	<u>3</u>	■ 1.000	b_i	b_i	vectors b_i . Training a DNN is the process of searching
BradleyGastaldiTerilla2023 F00069	<u>3</u>	■ 0.942	D	D	if one has a vocabulary D consisting of, say, 30,000 words,
BradleyGastaldiTerilla2024 F00070	<u>4</u>	■ 1.000	$D \rightarrow \mathbb{R}^D$	$D \rightarrow \mathbb{R}^D$	then $D \rightarrow \mathbb{R}^D$ embeds words as the standard basis vectors
BradleyGastaldiTerilla2024 F00071	<u>4</u>	■ 1.000	$\mathbb{R}^D$	$\mathbb{R}^D$	same $\mathbb{R}^D$, though it could certainly be different in applica-
BradleyGastaldiTerilla2024 F00072	<u>4</u>	■ 1.000	f_1	f_1	stance arose. If the first layer (namely, f_1 in Equation (1))
BradleyGastaldiTerilla2024 F00073	<u>4</u>	■ 1.000	σ	σ	single layer by σ , one can then obtain
BradleyGastaldiTerilla2024 F00074	<u>4</u>	■ 1.000	$\mathbb{R}^{n_1}$	$\mathbb{R}^{n_1}$	word vectors in $\mathbb{R}^{n_1}$ generally have nonzero entries.
BradleyGastaldiTerilla2024 F00075	<u>4</u>	■ 1.000	$\sigma: \mathbb{R}^D \rightarrow \mathbb{R}^{n_1}$	$\sigma: \mathbb{R}^D \rightarrow \mathbb{R}^{n_1}$	with the embedding map $\sigma: \mathbb{R}^D \rightarrow \mathbb{R}^{n_1}$ in (3) that was

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BradleyGastaldiTerilla2024 F00076	4	■ 0.998	$ D \times D $	$ D \times D $	alent to factorizing a $ D \times D $ matrix M , where the i - j -entry
BradleyGastaldiTerilla2024 F00077	4	■ 0.998	M	M	alent to factorizing a $ D \times D $ matrix M , where the i - j -entry
BradleyGastaldiTerilla2024 F00078	4	■ 0.998	i	i	alent to factorizing a $ D \times D $ matrix M , where the i - j -entry
BradleyGastaldiTerilla2024 F00079	4	■ 0.998	j	j	alent to factorizing a $ D \times D $ matrix M , where the i - j -entry
BradleyGastaldiTerilla2024 F00080	4	■ 1.000	$w_{\{i\}}$	w_i	sociation between words w_i and w_j . That measure is based
BradleyGastaldiTerilla2024 F00081	4	■ 1.000	$w_{\{j\}}$	w_j	sociation between words w_i and w_j . That measure is based
BradleyGastaldiTerilla2025 F00082	5	■ 1.000	$\sigma^{\{\prime\}}$	σ'	σ' and σ of sizes $ D \times d$ and $d \times D $ respectively, with
BradleyGastaldiTerilla2025 F00083	5	■ 1.000	$ D \times d$	$ D \times d$	σ' and σ of sizes $ D \times d$ and $d \times D $ respectively, with
BradleyGastaldiTerilla2025 F00084	5	■ 1.000	$d \times D $	$d \times D $	σ' and σ of sizes $ D \times d$ and $d \times D $ respectively, with
BradleyGastaldiTerilla2025 F00085	5	■ 1.000	$d \ll D $	$d \ll D $	$d \ll D $, such that $\ M - \sigma'\sigma\ $ is mimimal. The upshot is
BradleyGastaldiTerilla2025 F00086	5	■ 1.000	$\left M - \sigma^{\{\prime\}} \sigma\right $	$\ M - \sigma'\sigma\ $	$d \ll D $, such that $\ M - \sigma'\sigma\ $ is mimimal. The upshot is
BradleyGastaldiTerilla2025 F00087	5	■ 1.000	Y	Y	Given finite sets X and Y , an X - Y matrix valued in a
BradleyGastaldiTerilla2025 F00088	5	■ 1.000	X - Y	$X - Y$	Given finite sets X and Y , an X - Y matrix valued in a
BradleyGastaldiTerilla2025 F00089	5	■ 1.000	$m: X \times Y \rightarrow k$	$m: X \times Y \rightarrow k$	field k is a function $m: X \times Y \rightarrow k$. By simple curry-
BradleyGastaldiTerilla2025 F00090	5	■ 1.000	m	m	ing, m defines functions $X \rightarrow k^Y$ and $Y \rightarrow k^X$ defined by
BradleyGastaldiTerilla2025 F00091	5	■ 1.000	$X \rightarrow k^{\{Y\}}$	$X \rightarrow k^Y$	ing, m defines functions $X \rightarrow k^Y$ and $Y \rightarrow k^X$ defined by
BradleyGastaldiTerilla2025 F00092	5	■ 1.000	$Y \rightarrow k^{\{X\}}$	$Y \rightarrow k^X$	ing, m defines functions $X \rightarrow k^Y$ and $Y \rightarrow k^X$ defined by
BradleyGastaldiTerilla2025 F00093	5	■ 1.000	$x \mapsto m(x, -)$	$x \mapsto m(x, -)$	$x \mapsto m(x, -)$ and $y \mapsto m(-, y)$. Ordering the elements of X
BradleyGastaldiTerilla2025 F00094	5	■ 1.000	$y \mapsto m(-, y)$	$y \mapsto m(-, y)$	$x \mapsto m(x, -)$ and $y \mapsto m(-, y)$. Ordering the elements of X
BradleyGastaldiTerilla2025 F00095	5	■ 1.000	$ X $	$ X $	array of numbers with $ X $ rows and $ Y $ columns with the
BradleyGastaldiTerilla2025 F00096	5	■ 1.000	$ Y $	$ Y $	array of numbers with $ X $ rows and $ Y $ columns with the
BradleyGastaldiTerilla2025 F00097	5	■ 1.000	$m(x, y)$	$m(x, y)$	value $m(x, y)$ being the number in the x -th row and y -th
BradleyGastaldiTerilla2025 F00098	5	■ 1.000	$m(x, -) \in k^{\{Y\}}$	$m(x, -) \in k^Y$	column. The function $m(x, -) \in k^Y$ is then identified with
BradleyGastaldiTerilla2025 F00099	5	■ 1.000	$m(-, y) \in k^{\{X\}}$	$m(-, y) \in k^X$	m represents the function $m(-, y) \in k^X$. Linearly extend-
BradleyGastaldiTerilla2025 F00100	5	■ 1.000	$M^{\{*\}}: k^{\{X\}} \rightarrow k^{\{Y\}}$	$M^*: k^X \rightarrow k^Y$	$M^*: k^X \rightarrow k^Y$ and $M: k^Y \rightarrow k^X$, which of course are the
BradleyGastaldiTerilla2025 F00101	5	■ 1.000	$M: k^{\{Y\}} \rightarrow k^{\{X\}}$	$M: k^Y \rightarrow k^X$	$M^*: k^X \rightarrow k^Y$ and $M: k^Y \rightarrow k^X$, which of course are the
BradleyGastaldiTerilla2025 F00102	5	■ 0.998	$M^{\{*\}}$	M^*	M^* . Here is a diagram:

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BradleyGastaldiTerilla2023 F00103	5	1.000	$M M^*: k^X \rightarrow k^X$	$MM^*: k^X \rightarrow k^X$	Now, the compositions $MM^*: k^X \rightarrow k^X$ and
BradleyGastaldiTerilla2023 F00104	5	1.000	$M^* M: k^Y \rightarrow k^Y$	$M^*M: k^Y \rightarrow k^Y$	$M^*M: k^Y \rightarrow k^Y$ are linear operators with special prop-
BradleyGastaldiTerilla2023 F00105	5	1.000	$\mathbb{R}$	$\mathbb{R}$	bers $\mathbb{R}$, we can apply the spectral theorem to obtain or-
BradleyGastaldiTerilla2023 F00106	5	0.998	$\left\{u_1, \ldots, u_m\right\}$	$\{u_1, \ldots, u_m\}$	thonormal bases $\{u_1, \ldots, u_m\}$ of k^X and $\{v_1, \ldots, v_n\}$ of k^Y
BradleyGastaldiTerilla2023 F00107	5	0.998	$\left\{v_1, \ldots, v_n\right\}$	$\{v_1, \ldots, v_n\}$	thonormal bases $\{u_1, \ldots, u_m\}$ of k^X and $\{v_1, \ldots, v_n\}$ of k^Y
BradleyGastaldiTerilla2023 F00108	5	0.998	k^Y	k^Y	thonormal bases $\{u_1, \ldots, u_m\}$ of k^X and $\{v_1, \ldots, v_n\}$ of k^Y
BradleyGastaldiTerilla2023 F00109	5	1.000	$M M^*$	MM^*	consisting of eigenvectors of MM^* and M^*M respectively
BradleyGastaldiTerilla2023 F00110	5	1.000	$M^* M$	M^*M	consisting of eigenvectors of MM^* and M^*M respectively
BradleyGastaldiTerilla2023 F00111	5	1.000	$\left\{\lambda_1, \ldots, \lambda_r, 0, \ldots, 0\right\}$	$\{\lambda_1, \ldots, \lambda_r, 0, \ldots, 0\}$	with shared nonnegative real eigenvalues $\{\lambda_1, \ldots, \lambda_r, 0, \ldots, 0\}$,
BradleyGastaldiTerilla2023 F00112	5	1.000	$r=\min (m, n)$	$r = \min(m, n)$	where $r = \min(m, n)$. This data can be refashioned into a
BradleyGastaldiTerilla2023 F00113	5	1.000	$M=U \Sigma V^*$	$M = U\Sigma V^*$	factorization of M as $M = U\Sigma V^*$. This is the so-called sin-
BradleyGastaldiTerilla2023 F00114	5	1.000	$\left\{u_i\right\}$	$\{u_i\}$	gular value decomposition of M . The $\{u_i\}$ are the columns
BradleyGastaldiTerilla2023 F00115	5	0.998	U	U	of U , the $\{v_j\}$ are the rows of V^* , and Σ is the $m \times n$ diag-
BradleyGastaldiTerilla2023 F00116	5	0.998	$\left\{v_j\right\}$	$\{v_j\}$	of U , the $\{v_j\}$ are the rows of V^* , and Σ is the $m \times n$ diag-
BradleyGastaldiTerilla2023 F00117	5	0.998	V^*	V^*	of U , the $\{v_j\}$ are the rows of V^* , and Σ is the $m \times n$ diag-
BradleyGastaldiTerilla2023 F00118	5	0.998	Σ	Σ	of U , the $\{v_j\}$ are the rows of V^* , and Σ is the $m \times n$ diag-
BradleyGastaldiTerilla2023 F00119	5	0.998	$m \times n$	$m \times n$	of U , the $\{v_j\}$ are the rows of V^* , and Σ is the $m \times n$ diag-
BradleyGastaldiTerilla2023 F00120	5	1.000	$\sigma_i=\sqrt{\lambda_i}$	$\sigma_i = \sqrt{\lambda_i}$	onal matrix whose i -th entry is $\sigma_i = \sqrt{\lambda_i}$. The matrices U
BradleyGastaldiTerilla2023 F00121	5	1.000	V	V	and V satisfy $U^*U = I$ and $V^*V = I$. In SVD terminology,
BradleyGastaldiTerilla2023 F00122	5	1.000	$U^* U=I$	$U^*U = I$	and V satisfy $U^*U = I$ and $V^*V = I$. In SVD terminology,
BradleyGastaldiTerilla2023 F00123	5	1.000	$V^* V=I$	$V^*V = I$	and V satisfy $U^*U = I$ and $V^*V = I$. In SVD terminology,
BradleyGastaldiTerilla2023 F00124	5	1.000	σ_i	σ_i	the nonnegative real numbers σ_i are the <i>singular values</i> of
BradleyGastaldiTerilla2023 F00125	5	1.000	$\left\{u_1, \ldots, u_r\right\}$	$\{u_1, \ldots, u_r\}$	M , and the vectors $\{u_1, \ldots, u_r\}$ and $\{v_1, \ldots, v_r\}$ are the <i>left and</i>
BradleyGastaldiTerilla2023 F00126	5	1.000	$\left\{v_1, \ldots, v_r\right\}$	$\{v_1, \ldots, v_r\}$	M , and the vectors $\{u_1, \ldots, u_r\}$ and $\{v_1, \ldots, v_r\}$ are the <i>left and</i>
BradleyGastaldiTerilla2023 F00127	5	0.999	$\left\{\left(u_1, v_1\right), \ldots,\left(u_r, v_r\right)\right\}$	$\{(u_1, v_1), \ldots, (u_r, v_r)\}$	vectors $\{(u_1, v_1), \ldots, (u_r, v_r)\}$ in $k^X \times k^Y$ related to each other
BradleyGastaldiTerilla2023 F00128	5	0.999	$k^X \times k^Y$	$k^X \times k^Y$	vectors $\{(u_1, v_1), \ldots, (u_r, v_r)\}$ in $k^X \times k^Y$ related to each other
BradleyGastaldiTerilla2023 F00129	5	1.000	$\left(u_i, v_i\right) \leq\left(u_j, v_j\right)$	$(u_i, v_i) \leq (u_j, v_j)$	Moreover, these pairs are ordered with $\boxed{u_i, v_i} \leq \boxed{u_j, v_j}$ if

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BradleyGastaldiTerilla2023 F00130	5	■ 1.000	$\sigma_i \leq \sigma_j$	$\sigma_i \leq \sigma_j$	the corresponding singular values satisfy $\sigma_i \leq \sigma_j$. Finally,
BradleyGastaldiTerilla2023 F00131	5	■ 1.000	$M^{\prime}$	M'	it is not difficult to show that the matrix M' with rank at
BradleyGastaldiTerilla2023 F00132	5	■ 1.000	s	s	most s that is closest in Frobenius norm to the matrix M
BradleyGastaldiTerilla2023 F00133	5	■ 1.000	$M^{\prime}=U \Sigma^{\prime} V^{*}$	$M' = U \Sigma' V^*$	is given by $M' = U \Sigma' V^*$ where Σ' is the $m \times n$ diagonal
BradleyGastaldiTerilla2023 F00134	5	■ 1.000	$\Sigma^{\prime}$	Σ'	is given by $M' = U \Sigma' V^*$ where Σ' is the $m \times n$ diagonal
BradleyGastaldiTerilla2023 F00135	5	■ 1.000	$U, \Sigma^{\prime}$	U, Σ'	ues on the diagonal. By eliminating the parts of U, Σ' , and
BradleyGastaldiTerilla2023 F00136	5	■ 1.000	$U \Sigma^{\prime} V^{*}$	$U \Sigma' V^*$	ticipate in the product $U \Sigma' V^*$, one obtains a factorization
BradleyGastaldiTerilla2023 F00137	5	■ 0.998	$M^{\prime}=U^{\prime} \Sigma^{\prime} V^{\prime *} \approx M$	$M' = U' \Sigma'' V'^* \approx M$	$M' = U' \Sigma'' V'^* \approx M$, where U' is an $m \times s$ matrix, V'^* is
BradleyGastaldiTerilla2023 F00138	5	■ 0.998	$U^{\prime}$	U'	$M' = U' \Sigma'' V'^* \approx M$, where U' is an $m \times s$ matrix, V'^* is
BradleyGastaldiTerilla2023 F00139	5	■ 0.998	$m \times s$	$m \times s$	$M' = U' \Sigma'' V'^* \approx M$, where U' is an $m \times s$ matrix, V'^* is
BradleyGastaldiTerilla2023 F00140	5	■ 0.998	$V^{\prime *}$	V'^*	$M' = U' \Sigma'' V'^* \approx M$, where U' is an $m \times s$ matrix, V'^* is
BradleyGastaldiTerilla2023 F00141	5	■ 1.000	$s \times n$	$s \times n$	an $s \times n$ matrix, and Σ'' is an $s \times s$ diagonal matrix with
BradleyGastaldiTerilla2023 F00142	5	■ 1.000	$\Sigma^{\prime \prime}$	Σ''	an $s \times n$ matrix, and Σ'' is an $s \times s$ diagonal matrix with
BradleyGastaldiTerilla2023 F00143	5	■ 1.000	$s \times s$	$s \times s$	an $s \times n$ matrix, and Σ'' is an $s \times s$ diagonal matrix with
BradleyGastaldiTerilla2023 F00144	6	■ 0.855	$f: \mathrm{C}^{\mathrm{op}} \times \mathrm{D} \rightarrow$	$f: C^{\mathrm{op}} \times D \rightarrow$	<i>profunctor</i> , which is a set-valued functor $f: C^{\mathrm{op}} \times D \rightarrow \mathrm{Set}$.
BradleyGastaldiTerilla2023 F00145	6	■ 0.871	$\mathrm{C} \rightarrow \left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$	$C \rightarrow \left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$	rying, a profunctor defines a pair of functors $C \rightarrow \left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$
BradleyGastaldiTerilla2023 F00146	6	■ 0.592	$\mathrm{D} \rightarrow$	$D \rightarrow$	and $D \rightarrow \mathrm{Set}^{\mathrm{cop}}$ defined on objects by $c \mapsto f(c, -)$ and
BradleyGastaldiTerilla2023 F00147	6	■ 0.592	C^{op}	C^{op}	and $D \rightarrow \mathrm{Set}^{\mathrm{cop}}$ defined on objects by $c \mapsto f(c, -)$ and
BradleyGastaldiTerilla2023 F00148	6	■ 0.592	$c \mapsto f(c, -)$	$c \mapsto f(c, -)$	and $D \rightarrow \mathrm{Set}^{\mathrm{cop}}$ defined on objects by $c \mapsto f(c, -)$ and
BradleyGastaldiTerilla2023 F00149	6	■ 1.000	$d \mapsto f(-, d)$	$d \mapsto f(-, d)$	$d \mapsto f(-, d)$. As in the linear algebra setting, the functor
BradleyGastaldiTerilla2023 F00150	6	■ 1.000	$f(c, -)$	$f(c, -)$	$f(c, -)$ can be pictured as the c -th row of sets in the matrix
BradleyGastaldiTerilla2023 F00151	6	■ 1.000	c	c	$f(c, -)$ can be pictured as the c -th row of sets in the matrix
BradleyGastaldiTerilla2023 F00152	6	■ 1.000	$f(-, d): \mathrm{C}^{\mathrm{op}} \rightarrow$	$f(-, d): C^{\mathrm{op}} \rightarrow$	$f(-, d): C^{\mathrm{op}} \rightarrow \mathrm{Set}$ can be similarly be pictured as the d -th
BradleyGastaldiTerilla2023 F00153	6	■ 1.000	d	d	$f(-, d): C^{\mathrm{op}} \rightarrow \mathrm{Set}$ can be similarly be pictured as the d -th
BradleyGastaldiTerilla2023 F00154	6	■ 0.260	$\mathrm{C} \rightarrow \left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$	$C \rightarrow \left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$	ding, the functors $C \rightarrow \left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$ and $D \rightarrow \mathrm{Set}^{\mathrm{cop}}$ can be ex-
BradleyGastaldiTerilla2023 F00155	6	■ 0.260	C^{op}	C^{op}	ding, the functors $C \rightarrow \left(\mathrm{Set}^{\mathrm{D}}\right)^{\mathrm{op}}$ and $D \rightarrow \mathrm{Set}^{\mathrm{cop}}$ can be ex-

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BradleyGastaldiTerilla2015 F00156	6	■ 0.239	$F^{\{*\}}$	F^*	tended in a unique way to functors $F^* : \text{Set}^{\text{C}^{\text{op}}} \rightarrow (\text{Set}^{\text{D}})^{\text{op}}$
BradleyGastaldiTerilla2015 F00157	6	■ 0.239	$\{ \}^{\{\text{C}^{\text{op}}\}} \rightarrow \left(\text{Set}^{\text{D}} \right)^{\text{op}}$	$\text{C}^{\text{op}} \rightarrow \left(\text{Set}^{\text{D}} \right)^{\text{op}}$	tended in a unique way to functors $F^* : \text{Set}^{\text{C}^{\text{op}}} \rightarrow (\text{Set}^{\text{D}})^{\text{op}}$
BradleyGastaldiTerilla2015 F00158	6	■ 0.390	$F_{\{*\}} : \left(\text{Set}^{\{\text{D}\}} \right)^{\{\text{op}\}} \rightarrow$	$F_* : \left(\text{Set}^{\text{D}} \right)^{\text{op}} \rightarrow$	and $F_* : (\text{Set}^{\text{D}})^{\text{op}} \rightarrow \text{Set}^{\text{C}^{\text{op}}}$ that preserve colimits and lim-
BradleyGastaldiTerilla2015 F00159	6	■ 0.390	$\left(\text{C}^{\{\text{op}\}} \right)^{\{\text{op}\}}$	C^{op}	and $F_* : (\text{Set}^{\text{D}})^{\text{op}} \rightarrow \text{Set}^{\text{C}^{\text{op}}}$ that preserve colimits and lim-
BradleyGastaldiTerilla2015 F00160	6	■ 1.000	$F_{\{*\}}$	F_*	ties, the functors F^* and F_* are adjoint functors with spe-
BradleyGastaldiTerilla2015 F00161	6	■ 0.463	$\{ \}^{\{\text{D}\}}$	D	$(\text{Set}^{\text{D}})^{\text{op}} : F_*$ is known as the <i>Isbell adjunction</i> , which John
BradleyGastaldiTerilla2015 F00162	6	■ 1.000	$F^{\{*\}} F_{\{*\}}$	$F^* F_*$	functors $F^* F_*$ and $F_* F^*$ are called the <i>nuclei</i> of the profunc-
BradleyGastaldiTerilla2015 F00163	6	■ 1.000	$F_{\{*\}} F^{\{*\}}$	$F_* F^*$	functors $F^* F_*$ and $F_* F^*$ are called the <i>nuclei</i> of the profunc-
BradleyGastaldiTerilla2015 F00164	6	■ 0.974	$c_{\{i\}}, d_{\{i\}}$	c_i, d_i	(c_i, d_i) of objects in $\text{C}^{\text{op}} \times \text{D}$, where
BradleyGastaldiTerilla2015 F00165	6	■ 0.974	$\text{C}^{\{\text{op}\}} \times \text{D}$	$\text{C}^{\text{op}} \times \text{D}$	(c_i, d_i) of objects in $\text{C}^{\text{op}} \times \text{D}$, where
BradleyGastaldiTerilla2015 F00166	6	■ 0.999	$\left\{ \left(c_{\{i\}}, d_{\{i\}} \right) \right\}$	$\{(c_i, d_i)\}$	The nuclei themselves $\{(c_i, d_i)\}$ have significant structure —
BradleyGastaldiTerilla2015 F00167	6	■ 1.000	$\left\{ \left(u_{\{i\}}, v_{\{i\}} \right) \right\}$	$\{(u_i, v_i)\}$	plete. The pairs $\{(u_i, v_i)\}$ of singular vectors of a matrix
BradleyGastaldiTerilla2015 F00168	6	■ 1.000	r	r	of Set , then a profunctor r between two finite sets X and
BradleyGastaldiTerilla2015 F00169	6	■ 1.000	$r : X \times Y \rightarrow \{0, 1\}$	$r : X \times Y \rightarrow \{0, 1\}$	function $r : X \times Y \rightarrow \{0, 1\}$, which is the same as a relation
BradleyGastaldiTerilla2015 F00170	6	■ 1.000	$X \times Y$	$X \times Y$	on $X \times Y$. The functors $R^* : 2^X \rightarrow 2^Y$ and $R_* : 2^Y \rightarrow 2^X$
BradleyGastaldiTerilla2015 F00171	6	■ 1.000	$R^{\{*\}} : 2^{\{X\}} \rightarrow 2^{\{Y\}}$	$R^* : 2^X \rightarrow 2^Y$	on $X \times Y$. The functors $R^* : 2^X \rightarrow 2^Y$ and $R_* : 2^Y \rightarrow 2^X$
BradleyGastaldiTerilla2015 F00172	6	■ 1.000	$R_{\{*\}} : 2^{\{Y\}} \rightarrow 2^{\{X\}}$	$R_* : 2^Y \rightarrow 2^X$	on $X \times Y$. The functors $R^* : 2^X \rightarrow 2^Y$ and $R_* : 2^Y \rightarrow 2^X$
BradleyGastaldiTerilla2015 F00173	6	■ 1.000	$R^{\{*\}}$	R^*	[GW99]. The function R^* maps a subset $A \subseteq X$ to the set
BradleyGastaldiTerilla2015 F00174	6	■ 1.000	$A \subseteq X$	$A \subseteq X$	[GW99]. The function R^* maps a subset $A \subseteq X$ to the set
BradleyGastaldiTerilla2015 F00175	6	■ 1.000	$R^{\{*\}}(A) = \{y \in Y : R(x, y) = 1\}$	$R^*(A) = \{y \in Y : R(x, y) = 1\}$	$R^*(A) = \{y \in Y : R(x, y) = 1\}$ for all $x \in A$ and R_* maps
BradleyGastaldiTerilla2015 F00176	6	■ 1.000	$x \in A$	$x \in A$	$R^*(A) = \{y \in Y : R(x, y) = 1 \text{ for all } x \in A\}$ and R_* maps
BradleyGastaldiTerilla2015 F00177	6	■ 1.000	$R_{\{*\}}$	R_*	$R^*(A) = \{y \in Y : R(x, y) = 1 \text{ for all } x \in A\}$ and R_* maps
BradleyGastaldiTerilla2015 F00178	6	■ 1.000	$B \subseteq Y$	$B \subseteq Y$	a subset $B \subseteq Y$ to the set $R_*(B) = \{x \in X : R(x, y) =$
BradleyGastaldiTerilla2015 F00179	6	■ 1.000	$R_{\{*\}}(B) = \{x \in X : R(x, y) =$	$R_*(B) = \{x \in X : R(x, y) =$	a subset $B \subseteq Y$ to the set $R_*(B) = \{x \in X : R(x, y) =$
BradleyGastaldiTerilla2015 F00180	6	■ 1.000	$y \in B\}$	$y \in B\}$	1 for all $y \in B\}$. The fixed objects of $R^* R_*$ and $R_* R^*$ are
BradleyGastaldiTerilla2015 F00181	6	■ 1.000	$R^{\{*\}} R_{\{*\}}$	$R^* R_*$	1 for all $y \in B\}$. The fixed objects of $R^* R_*$ and $R_* R^*$ are
BradleyGastaldiTerilla2015 F00182	6	■ 1.000	$R_{\{*\}} R^{\{*\}}$	$R_* R^*$	1 for all $y \in B\}$. The fixed objects of $R^* R_*$ and $R_* R^*$ are

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BradleyGastaldiTerilla2021 F00183	6	■ 0.989	$\left(A_{\{i\}}, B_{\{i\}}\right) \subset X \times Y$	$(A_i, B_i) \subset X \times Y$	$(A_i, B_i) \subset X \times Y$ with
BradleyGastaldiTerilla2021 F00184	6	■ 1.000	$\left\{\left(\left(A_{\{i\}}, B_{\{i\}}\right)\right)\right\}$	$\{(A_i, B_i)\}$	and the set of all formal concepts $\{(A_i, B_i)\}$ is partially or-
BradleyGastaldiTerilla2021 F00185	6	■ 0.915	$\left(A_{\{i\}}, B_{\{i\}}\right) \leq \left(A_{\{j\}}, B_{\{j\}}\right)$	$(A_i, B_i) \leq (A_j, B_j)$	dered with $(A_i, B_i) \leq (A_j, B_j)$ if and only if $A_i \subseteq A_j$ which
BradleyGastaldiTerilla2021 F00186	6	■ 0.915	$A_{\{i\}} \subseteq A_{\{j\}}$	$A_i \subseteq A_j$	dered with $(A_i, B_i) \leq (A_j, B_j)$ if and only if $A_i \subseteq A_j$ which
BradleyGastaldiTerilla2021 F00187	6	■ 1.000	$B_{\{i\}} \supseteq B_{\{j\}}$	$B_i \supseteq B_j$	is equivalent to $B_i \supseteq B_j$. Moreover, $\{(A_i, B_i)\}$ forms a com-
BradleyGastaldiTerilla2021 F00188	6	■ 0.964	$\left(A_{\{i\}}, B_{\{j\}}\right) \wedge \left(A_{\{j\}}, B_{\{j\}}\right) := \left(A_{\{i\}} \cap A_{\{j\}}, R^* R_* \left(B_i \cup B_j\right)\right)$	$(A_i, B_j) \wedge (A_j, B_j) := (A_i \cap A_j, R^* R_* (B_i \cup B_j))$	finied by $(A_i, B_i) \wedge (A_j, B_j) := (A_i \cap A_j, R^* R_* (B_i \cup B_j))$ and
BradleyGastaldiTerilla2021 F00189	6	■ 1.000	$\left(A_{\{i\}}, B_{\{j\}}\right) \vee \left(A_{\{j\}}, B_{\{j\}}\right) := \left(R_* R^* \left(A_i \cup A_j\right), B_i \cap B_j\right)$	$(A_i, B_j) \vee (A_j, B_j) := (R_* R^* (A_i \cup A_j), B_i \cap B_j)$	$(A_i, B_i) \vee (A_j, B_j) := (R_* R^* (A_i \cup A_j), B_i \cap B_j)$. The point is
BradleyGastaldiTerilla2021 F00190	7	■ 1.000	$Y = X \times X$	$Y = X \times X$	Now let $Y = X \times X$ and, in line with what we have seen in
BradleyGastaldiTerilla2021 F00191	7	■ 1.000	$m: X \times Y \rightarrow \mathbb{R}$	$m: X \times Y \rightarrow \mathbb{R}$	the previous sections, consider a matrix $m: X \times Y \rightarrow \mathbb{R}$
BradleyGastaldiTerilla2021 F00192	7	■ 1.000	$\left(y_{\{l\}}, y_{\{r\}}\right) \in Y$	$(y_l, y_r) \in Y$	characters $(y_l, y_r) \in Y$ are the left and right contexts of the
BradleyGastaldiTerilla2021 F00193	7	■ 1.000	$m(\mathrm{h},(\mathrm{t}, \mathrm{e})) \approx$	$m(\mathrm{h},(\mathrm{t}, \mathrm{e})) \approx$	character $x \in X$. For instance, we have that $m(\mathrm{h},(\mathrm{t}, \mathrm{e})) \approx$
BradleyGastaldiTerilla2021 F00194	7	■ 1.000	$m(\mathrm{h},(\mathrm{p}, \mathrm{o})) \approx 0.0037$	$m(\mathrm{h},(\mathrm{p}, \mathrm{o})) \approx 0.0037$	0.3836 while $m(\mathrm{h},(\mathrm{p}, \mathrm{o})) \approx 0.0037$ reflecting that it is over
BradleyGastaldiTerilla2021 F00195	7	■ 1.000	$X \rightarrow \mathbb{R}^Y$	$X \rightarrow \mathbb{R}^Y$	ever, embedding them $X \rightarrow \mathbb{R}^Y$ via the matrix m and lever-
BradleyGastaldiTerilla2021 F00196	7	■ 1.000	$M^*: \mathbb{R}^X \rightarrow \mathbb{R}^Y$	$M^*: \mathbb{R}^X \rightarrow \mathbb{R}^Y$	duced operator $M^*: \mathbb{R}^X \rightarrow \mathbb{R}^Y$ we can obtain a vector rep-
BradleyGastaldiTerilla2021 F00197	8	■ 1.000	$\{(A, B)\}$	$\{(A, B)\}$	concepts $\{(A, B)\}$ for which A contains the selected char-
BradleyGastaldiTerilla2021 F00198	8	■ 1.000	A	A	concepts $\{(A, B)\}$ for which A contains the selected char-
BradleyGastaldiTerilla2021 F00199	8	■ 1.000	$ B \geq 20$	$ B \geq 20$	nodes representing a large number of contexts, $ B \geq 20$,
BradleyGastaldiTerilla2021 F00200	8	■ 1.000	$A_{\{i\}}$	A_i	of characters A_i is associated to an explicit dual set of
BradleyGastaldiTerilla2021 F00201	8	■ 1.000	$B_{\{i\}}$	B_i	contexts B_i suggesting a principle of compositionality —
BradleyGastaldiTerilla2021 F00202	8	■ 1.000	$\vee$	$\vee$	aforementioned operations $\vee$ and $\wedge$ allowing one to per-
BradleyGastaldiTerilla2021 F00203	8	■ 1.000	$\wedge$	$\wedge$	aforementioned operations $\vee$ and $\wedge$ allowing one to per-
BradleyGastaldiTerilla2021 F00204	8	■ 1.000	$y_{\{l\}}, y_{\{r\}}$	y_l, y_r	Use the empirical probabilities that the words y_l, y_r are the
BradleyGastaldiTerilla2021 F00205	11	■ 1.000	$[-\infty, \infty]$	$[-\infty, \infty]$	closer together. The extended real line $[-\infty, \infty]$ or the unit
BradleyGastaldiTerilla2021 F00206	11	■ 1.000	$[-\infty, \infty]^X$	$[-\infty, \infty]^X$	$[-\infty, \infty]^X$, the functions on a set X valued in the extended
BradleyGastaldiTerilla2021 F00207	11	■ 1.000	2^X	2^X	over $[-\infty, \infty]$, much in the same way that 2^X can be viewed
BradleyGastaldiTerilla2021 F00208	11	■ 1.000	$\{0 \rightarrow 1\}$	$\{0 \rightarrow 1\}$	as a category of presheaves enriched over $\{0 \rightarrow 1\}$. Then,

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BradleyGastaldiTerilla2021 F00209	11	■ 1.000	$V_{\{1\}} \otimes V_{\{2\}} \otimes \cdots \otimes V_{\{n\}}$	$V_1 \otimes V_2 \otimes \cdots \otimes V_n$	 sor in the tensor product of vector spaces
BradleyGastaldiTerilla2021 F00210	11	■ 1.000	$n-1$	$n - 1$	can be interpreted as a sequence of $n - 1$ compatible trun-
BradleyGastaldiTerilla2021 F00211	11	■ 0.996	$\mathrm{C}_{\{1\}} \times \mathrm{C}_{\{2\}} \times \cdots \times \mathrm{C}_{\{n\}}$	$C_1 \times C_2 \times \cdots \times C_n$	of categories $C_1 \times C_2 \times \cdots \times C_n$. Given that text data is more